Assignment 1

1] Proof that Eqn. 1 is derivative of Eqn. 2

$$q(t) = \sum_{k=0}^{\infty} \frac{(\hat{\omega} t)^k}{k!} q(0)$$

$$= \left(1 + \hat{\omega} t + \frac{(\hat{\omega} t)^2}{2!} + \dots + \frac{(\hat{\omega} t)^k}{k!} + \dots \right) q(0)$$

$$\dot{q}(t) = \left(0 + \hat{\omega} + \frac{2 \hat{\omega}^2 t}{2!} + \dots + \frac{k \hat{\omega}^k t^{k-1}}{k!} + \dots \right) q(0) \qquad \| \frac{k}{k!} = \frac{1}{(k-1)!}$$

$$= \hat{\omega} \left(1 + \hat{\omega} t + \dots + \frac{(\hat{\omega} t)^{k-1}}{(k-1)!} + \dots \right) q(0) \qquad \| \text{Let } m = k-1$$

$$= \hat{\omega} \sum_{m=0}^{\infty} \frac{(\hat{\omega} t)^m}{m!} q(0) = \underline{\underline{\hat{\omega} q(t)}} \qquad \square$$

2] Show $\underline{\hat{\omega}^2 = \omega \omega^T - \|\omega\|^2 Id_3}$

$$\longrightarrow \hat{\omega}^2 = (1 - 1^T)^2$$

$$= \left(\begin{bmatrix} 0 & 0 & 0 \\ \omega_3 & 0 & 0 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} - \begin{bmatrix} 0 & \omega_3 & -\omega_2 \\ 0 & 0 & \omega_1 \\ 0 & 0 & 0 \end{bmatrix} \right)^2$$

$$= \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}$$

$$= \underline{\underline{\begin{bmatrix} -\omega_2^2 - \omega_3^2 & \omega_1 \omega_2 & \omega_1 \omega_3 \\ \omega_1 \omega_2 & -\omega_1^2 - \omega_3^2 & \omega_2 \omega_3 \\ \omega_1 \omega_3 & \omega_2 \omega_3 & -\omega_1^2 - \omega_2^2 \end{bmatrix}}}$$

$$\longrightarrow \omega \omega^T = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \cdot \begin{bmatrix} \omega_1 & \omega_2 & \omega_3 \end{bmatrix} = \underline{\begin{bmatrix} \omega_1^2 & \omega_1 \omega_2 & \omega_1 \omega_3 \\ \omega_1 \omega_2 & \omega_2^2 & \omega_2 \omega_3 \\ \omega_1 \omega_3 & \omega_2 \omega_3 & \omega_3^2 \end{bmatrix}}$$

$$\longrightarrow \|\omega\|^2 Id_3 = \omega_1^2 + \omega_2^2 + \omega_3^2 \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \underline{\begin{bmatrix} \omega_1^2 + \omega_2^2 + \omega_3^2 & 0 & 0 \\ 0 & \omega_1^2 + \omega_2^2 + \omega_3^2 & 0 \\ 0 & 0 & \omega_1^2 + \omega_2^2 + \omega_3^2 \end{bmatrix}}$$

$$\hookrightarrow \omega\omega^T - \|\omega\|^2 \mathrm{Id}_3 = \begin{bmatrix} \omega_1^2 & \omega_1\omega_2 & \omega_1\omega_3 \\ \omega_1\omega_2 & \omega_2^2 & \omega_2\omega_3 \\ \omega_1\omega_3 & \omega_2\omega_3 & \omega_3^2 \end{bmatrix} - \begin{bmatrix} \omega_1^2+\omega_2^2+\omega_3^2 & 0 & 0 \\ 0 & \omega_1^2+\omega_2^2+\omega_3^2 & 0 \\ 0 & 0 & \omega_1^2+\omega_2^2+\omega_3^2 \end{bmatrix}$$

$$= \begin{bmatrix} -\omega_2^2-\omega_3^2 & \omega_1\omega_2 & \omega_1\omega_3 \\ \omega_1\omega_2 & -\omega_1^2-\omega_3^2 & \omega_2\omega_3 \\ \omega_1\omega_3 & \omega_2\omega_3 & -\omega_1^2-\omega_2^2 \end{bmatrix} = \underline{\underline{\hat{\omega}^2}} \quad \square$$

Show $\underline{\hat{\omega}^3 = -\|\omega\|^2 \hat{\omega}}$

$$\longrightarrow \hat{\omega}\omega = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \cdot \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = \begin{bmatrix} \omega_2\omega_3 - \omega_2\omega_3 \\ \omega_1\omega_3 - \omega_1\omega_3 \\ \omega_1\omega_2 - \omega_1\omega_2 \end{bmatrix} = \underline{0}$$

$$\hookrightarrow \hat{\omega}^3 = \hat{\omega}\hat{\omega}^2 = \hat{\omega}\left(\omega\omega^T - \|\omega\|^2 \mathrm{Id}_3\right)$$

$$= \hat{\omega}\omega\omega^T - \|\omega\|^2 \hat{\omega}\,\mathrm{Id}_3 \qquad \| \ \hat{\omega}\omega = 0$$

$$= 0 - \|\omega\|^2 \hat{\omega}\,\mathrm{Id}_3 \qquad \| \ \hat{\omega}\,\mathrm{Id}_3 = \hat{\omega}$$

$$= \underline{\underline{-\|\omega\|^2 \hat{\omega}}} \quad \square$$

3] Show $\underline{R(\omega,\theta) = \exp(\hat{\omega}\theta) = \mathrm{Id}_3 + \hat{\omega}\sin(\theta) + \hat{\omega}^2(1-\cos(\theta))}$

$\longrightarrow$ Let $\|\omega\| = 1$. As $\hat{\omega}^3 = -\|\omega\|^2\hat{\omega} = -\hat{\omega}$, $\hat{\omega}^n$ follows the periodic pattern:

$\longrightarrow$ Odd case $(n = 2k+1)$: $\hat{\omega}^1 = \hat{\omega}$, $\hat{\omega}^3 = -\hat{\omega}$, $\hat{\omega}^5 = \hat{\omega}$, $\ldots \Rightarrow \underline{\hat{\omega}^{2k+1} = (-1)^k \hat{\omega}}$

$\longrightarrow$ Even case $(n = 2k)$: $\hat{\omega}^2 = \hat{\omega}^2$, $\hat{\omega}^4 = -\hat{\omega}^2$, $\hat{\omega}^6 = \hat{\omega}^2$, $\ldots \Rightarrow \underline{\hat{\omega}^{2k} = (-1)^{k-1} \hat{\omega}^2}$

$$R(\omega,\theta) = \exp(\hat{\omega}\theta)$$

$$= \sum_{n=0}^{\infty} \frac{(\hat{\omega}\theta)^n}{n!}$$

$$= \mathrm{Id}_3 + \sum_{n=1}^{\infty} \frac{\theta^n}{n!}\hat{\omega}^n \qquad \left\| \begin{array}{l} \text{Split into odd} \\ \text{and even terms} \end{array} \right.$$

$$= \mathrm{Id}_3 + \sum_{k=0}^{\infty} \frac{\theta^{2k+1}}{(2k+1)!}(-1)^k \hat{\omega} + \sum_{k=1}^{\infty} \frac{\theta^{2k}}{(2k)!}(-1)^{k-1}\hat{\omega}^2$$

$$= Id_3 + \hat{\omega} \sum_{k=0}^{\infty} (-1)^k \frac{\theta^{2k+1}}{(2k+1)!} + \hat{\omega}^2 \left(\sum_{k=1}^{\infty} (-1)^{k-1} \frac{\theta^{2k}}{(2k)!} \right)$$ // Def sin, // $(-1)^{k-1} = -(-1)^k$

$$= Id_3 + \hat{\omega} \sin(\theta) + \hat{\omega}^2 \left(-\sum_{k=1}^{\infty} (-1)^k \frac{\theta^{2k}}{(2k)!} \right)$$

$$= Id_3 + \hat{\omega} \sin(\theta) + \hat{\omega}^2 \left((-1)^0 \frac{\theta^{2\cdot 0}}{(2\cdot 0)!} - \sum_{k=0}^{\infty} (-1)^k \frac{\theta^{2k}}{(2k)!} \right)$$ // Def cos

$$\underline{\underline{= Id_3 + \hat{\omega} \sin(\theta) + \hat{\omega}^2 \left(1 - \cos(\theta) \right)}} \qquad \square$$

CA_sheet_2_assignment_2

May 16, 2026

Assignment 2 (Linear Interpolation of Quaternions and Euler Angles, 3 + 3 = 6 Points) `author: Rofikul Masud s82rmasu`

[16]:
```
import numpy as np
import pytest
from typing import Union
import subprocess
import sys
from IPython.display import display, HTML

try:
    import pytest
except ImportError:
    import subprocess
    subprocess.check_call([sys.executable, "-m", "pip", "install", "pytest",␣
↪"-q"])
    import pytest
```

Helper func from sheet 01

[17]:
```
# ------------------------------------------------------------------------------
# Conventions used in this notebook
# ------------------------------------------------------------------------------
# Quaternions are represented as [w, x, y, z].
# Euler angles are represented in the order specified by `order`.
# Example: order="ZYX" means euler=[z_angle, y_angle, x_angle], and
# R = Rz(z_angle) @ Ry(y_angle) @ Rx(x_angle).
# All six Tait-Bryan Euler orders are supported:
# "XYZ", "XZY", "YXZ", "YZX", "ZXY", "ZYX".
# ------------------------------------------------------------------------------
np.set_printoptions(precision=5, suppress=True)
EPS = 1e-10

ORDER = "ZYX"

def normalize_vector(v: np.ndarray, axis: int = -1, eps: float = EPS) -> np.
↪ndarray:
    """Normalize vectors along the given axis."""
```

```
    v = np.asarray(v, dtype=float)
    n = np.linalg.norm(v, axis=axis, keepdims=True)
    if np.any(n < eps):
        raise ValueError("Cannot normalize a vector with norm close to zero.")
    return v / n


def normalize_quaternion(q: np.ndarray, eps: float = EPS) -> np.ndarray:
    """Normalize one quaternion of shape (4,) or an array of quaternions (...,␣
 ↪4)."""
    q = np.asarray(q, dtype=float)
    if q.shape[-1] != 4:
        raise ValueError("Quaternions must have shape (..., 4).")
    n = np.linalg.norm(q, axis=-1, keepdims=True)
    if np.any(n < eps):
        raise ValueError("Cannot normalize a quaternion with norm close to zero.
 ↪")
    return q / n


def quaternion_conjugate(q: np.ndarray) -> np.ndarray:
    """Return the conjugate of q = [w, x, y, z]."""
    q = np.asarray(q, dtype=float)
    if q.shape[-1] != 4:
        raise ValueError("Quaternions must have shape (..., 4).")
    out = q.copy()
    out[..., 1:] *= -1.0
    return out


def quaternion_norm(q: np.ndarray) -> np.ndarray:
    """Return the Euclidean norm of one quaternion or a batch of quaternions."""
    q = np.asarray(q, dtype=float)
    if q.shape[-1] != 4:
        raise ValueError("Quaternions must have shape (..., 4).")
    return np.linalg.norm(q, axis=-1)


def quaternion_multiply(q_left: np.ndarray, q_right: np.ndarray) -> np.ndarray:
    """Hamilton product q_left * q_right for quaternions [w, x, y, z]."""
    q_left = np.asarray(q_left, dtype=float)
    q_right = np.asarray(q_right, dtype=float)
    if q_left.shape[-1] != 4 or q_right.shape[-1] != 4:
        raise ValueError("Both inputs must have shape (..., 4).")

    w1, x1, y1, z1 = np.moveaxis(q_left, -1, 0)
    w2, x2, y2, z2 = np.moveaxis(q_right, -1, 0)
```

```
    return np.stack([
        w1*w2 - x1*x2 - y1*y2 - z1*z2,
        w1*x2 + x1*w2 + y1*z2 - z1*y2,
        w1*y2 - x1*z2 + y1*w2 + z1*x2,
        w1*z2 + x1*y2 - y1*x2 + z1*w2,
    ], axis=-1)


def rotation_matrix_x(angle: float) -> np.ndarray:
    """Rotation matrix for a rotation about the x-axis."""
    c, s = np.cos(angle), np.sin(angle)
    return np.array([[1.0, 0.0, 0.0], [0.0, c, -s], [0.0, s, c]])


def rotation_matrix_y(angle: float) -> np.ndarray:
    """Rotation matrix for a rotation about the y-axis."""
    c, s = np.cos(angle), np.sin(angle)
    return np.array([[c, 0.0, s], [0.0, 1.0, 0.0], [-s, 0.0, c]])


def rotation_matrix_z(angle: float) -> np.ndarray:
    """Rotation matrix for a rotation about the z-axis."""
    c, s = np.cos(angle), np.sin(angle)
    return np.array([[c, -s, 0.0], [s, c, 0.0], [0.0, 0.0, 1.0]])


def axis_rotation_matrix(axis: str, angle: float) -> np.ndarray:
    """Return a basic rotation matrix for the selected axis."""
    axis = axis.upper()
    if axis == "X":
        return rotation_matrix_x(angle)
    if axis == "Y":
        return rotation_matrix_y(angle)
    if axis == "Z":
        return rotation_matrix_z(angle)
    raise ValueError("axis must be 'X', 'Y', or 'Z'.")


def axis_angle_quaternion(axis: str, angle: float) -> np.ndarray:
    """Quaternion [w, x, y, z] for a rotation around one coordinate axis."""
    half = angle / 2.0
    c, s = np.cos(half), np.sin(half)
    axis = axis.upper()
    if axis == "X":
        return np.array([c, s, 0.0, 0.0])
    if axis == "Y":
```

```python
        return np.array([c, 0.0, s, 0.0])
    if axis == "Z":
        return np.array([c, 0.0, 0.0, s])
    raise ValueError("axis must be 'X', 'Y', or 'Z'.")


def euler_to_rotation_matrix(euler: np.ndarray, order: str = ORDER) -> np.
↪ndarray:
    """
    Convert Euler angles to a rotation matrix.

    The angles are interpreted in the same order as the order string.
    Example: order="ZYX", euler=[z_angle, y_angle, x_angle].
    """
    euler = np.asarray(euler, dtype=float).reshape(3,)

    R = np.eye(3)
    for axis, angle in zip(order, euler):
        R = R @ axis_rotation_matrix(axis, angle)
    return R


def euler_to_quaternion(euler: np.ndarray, order: str = ORDER) -> np.ndarray:
    """
    Convert Euler angles to a unit quaternion [w, x, y, z].

    The angles are interpreted in the same order as the order string.
    Example: order="ZYX", euler=[z_angle, y_angle, x_angle].
    """
    euler = np.asarray(euler, dtype=float).reshape(3,)

    q = np.array([1.0, 0.0, 0.0, 0.0])
    for axis, angle in zip(order, euler):
        q = quaternion_multiply(q, axis_angle_quaternion(axis, angle))
    return normalize_quaternion(q)


def quaternion_to_rotation_matrix(q: np.ndarray) -> np.ndarray:
    """Convert q = [w, x, y, z] to a 3 x 3 rotation matrix."""
    q = normalize_quaternion(np.asarray(q, dtype=float).reshape(4,))
    w, x, y, z = q
    return np.array([
        [1 - 2*(y*y + z*z), 2*(x*y - w*z),     2*(x*z + w*y)],
        [2*(x*y + w*z),     1 - 2*(x*x + z*z), 2*(y*z - w*x)],
        [2*(x*z - w*y),     2*(y*z + w*x),     1 - 2*(x*x + y*y)],
    ], dtype=float)
```

```python
def rotation_matrix_to_quaternion(R: np.ndarray) -> np.ndarray:
    """Convert a 3 x 3 rotation matrix to a unit quaternion [w, x, y, z]."""
    R = np.asarray(R, dtype=float).reshape(3, 3)
    trace = np.trace(R)

    if trace > 0.0:
        s = 2.0 * np.sqrt(trace + 1.0)
        w = 0.25 * s
        x = (R[2, 1] - R[1, 2]) / s
        y = (R[0, 2] - R[2, 0]) / s
        z = (R[1, 0] - R[0, 1]) / s
    elif R[0, 0] > R[1, 1] and R[0, 0] > R[2, 2]:
        s = 2.0 * np.sqrt(1.0 + R[0, 0] - R[1, 1] - R[2, 2])
        w = (R[2, 1] - R[1, 2]) / s
        x = 0.25 * s
        y = (R[0, 1] + R[1, 0]) / s
        z = (R[0, 2] + R[2, 0]) / s
    elif R[1, 1] > R[2, 2]:
        s = 2.0 * np.sqrt(1.0 + R[1, 1] - R[0, 0] - R[2, 2])
        w = (R[0, 2] - R[2, 0]) / s
        x = (R[0, 1] + R[1, 0]) / s
        y = 0.25 * s
        z = (R[1, 2] + R[2, 1]) / s
    else:
        s = 2.0 * np.sqrt(1.0 + R[2, 2] - R[0, 0] - R[1, 1])
        w = (R[1, 0] - R[0, 1]) / s
        x = (R[0, 2] + R[2, 0]) / s
        y = (R[1, 2] + R[2, 1]) / s
        z = 0.25 * s

    return normalize_quaternion(np.array([w, x, y, z]))


def rotation_matrix_to_euler(R: np.ndarray, order: str = ORDER, eps: float =␣
 ↪1e-8) -> np.ndarray:
    """
    Convert a rotation matrix to Euler angles for any of the six supported␣
 ↪orders.

    Returns angles in the same order as `order`.
    Example: order="ZYX" returns [z_angle, y_angle, x_angle].
    In gimbal-lock cases, the last angle is set to zero.
    """
    R = np.asarray(R, dtype=float).reshape(3, 3)

    if order == "XYZ":
```

```python
        sb = np.clip(R[0, 2], -1.0, 1.0)
        b = np.arcsin(sb)
        if abs(abs(sb) - 1.0) < eps:
            c = 0.0
            a = np.arctan2(R[1, 0], R[1, 1]) if sb > 0 else np.arctan2(-R[1,␣
↪0], R[1, 1])
        else:
            a = np.arctan2(-R[1, 2], R[2, 2])
            c = np.arctan2(-R[0, 1], R[0, 0])

    elif order == "XZY":
        sb = np.clip(-R[0, 1], -1.0, 1.0)
        b = np.arcsin(sb)
        if abs(abs(sb) - 1.0) < eps:
            c = 0.0
            a = np.arctan2(-R[1, 2], R[1, 0]) if sb > 0 else np.arctan2(-R[1,␣
↪2], -R[1, 0])
        else:
            a = np.arctan2(R[2, 1], R[1, 1])
            c = np.arctan2(R[0, 2], R[0, 0])

    elif order == "YXZ":
        sb = np.clip(-R[1, 2], -1.0, 1.0)
        b = np.arcsin(sb)
        if abs(abs(sb) - 1.0) < eps:
            c = 0.0
            a = np.arctan2(-R[2, 0], R[0, 0])
        else:
            a = np.arctan2(R[0, 2], R[2, 2])
            c = np.arctan2(R[1, 0], R[1, 1])

    elif order == "YZX":
        sb = np.clip(R[1, 0], -1.0, 1.0)
        b = np.arcsin(sb)
        if abs(abs(sb) - 1.0) < eps:
            c = 0.0
            a = np.arctan2(R[0, 2], R[2, 2])
        else:
            a = np.arctan2(-R[2, 0], R[0, 0])
            c = np.arctan2(-R[1, 2], R[1, 1])

    elif order == "ZXY":
        sb = np.clip(R[2, 1], -1.0, 1.0)
        b = np.arcsin(sb)
        if abs(abs(sb) - 1.0) < eps:
            c = 0.0
            a = np.arctan2(R[1, 0], R[0, 0])
```

```
        else:
            a = np.arctan2(-R[0, 1], R[1, 1])
            c = np.arctan2(-R[2, 0], R[2, 2])

    elif order == "ZYX":
        sb = np.clip(-R[2, 0], -1.0, 1.0)
        b = np.arcsin(sb)
        if abs(abs(sb) - 1.0) < eps:
            c = 0.0
            a = np.arctan2(R[1, 2], R[0, 2]) if sb > 0 else np.arctan2(-R[1,␣
↪2], -R[0, 2])
        else:
            a = np.arctan2(R[1, 0], R[0, 0])
            c = np.arctan2(R[2, 1], R[2, 2])

    return np.array([a, b, c])


def quaternion_to_euler(q: np.ndarray, order: str = ORDER) -> np.ndarray:
    """Convert a quaternion [w, x, y, z] to Euler angles in the requested order.
↪"""
    return rotation_matrix_to_euler(quaternion_to_rotation_matrix(q),␣
↪order=order)


def rotation_direction_from_matrix(R: np.ndarray) -> np.ndarray:
    """Direction of the local +x axis after applying R."""
    return R @ np.array([1.0, 0.0, 0.0])


def directions_from_quaternions(quaternions: np.ndarray) -> np.ndarray:
    """Map a sequence of quaternions to the corresponding local +x directions.
↪"""
    return np.
↪array([rotation_direction_from_matrix(quaternion_to_rotation_matrix(q)) for␣
↪q in quaternions])


def directions_from_eulers(eulers: np.ndarray, order: str = ORDER) -> np.
↪ndarray:
    """Map a sequence of Euler angles to the corresponding local +x directions.
↪"""
    return np.array([rotation_direction_from_matrix(euler_to_rotation_matrix(e,␣
↪order=order)) for e in eulers])
```

```python
def relative_rotation_angle(R0: np.ndarray, R1: np.ndarray) -> float:
    """Return the angle of the relative rotation R0.T @ R1 in radians."""
    R_rel = R0.T @ R1
    c = (np.trace(R_rel) - 1.0) / 2.0
    return float(np.arccos(np.clip(c, -1.0, 1.0)))
```

SLERP Quaternion Implementation

[18]:
```python
def slerp_quaternion(q_start: np.ndarray, q_end: np.ndarray, t: Union[np.
 ↪ndarray, float]) -> np.ndarray:
    """
    Note:
    Spherical Linear Interpolation (SLERP) between two quaternions.
     performs smooth interpolation along the shortest path on the unit␣
 ↪quaternion sphere.

    Args:
        q_start: Starting quaternion [w, x, y, z]
        q_end: Ending quaternion [w, x, y, z]
        t: Interpolation parameter(s), scalar or 1D array with values in [0, 1]␣
 ↪- interpolation factors? 0 = q_start, 1 = q_end

    Returns:
        Interpolated quaternions with same shape as t (scalar or array)
    """
    #  inputs are numpy arrays and normalize
    q_start = normalize_quaternion(np.asarray(q_start, dtype=float).reshape(4,))
    q_end = normalize_quaternion(np.asarray(q_end, dtype=float).reshape(4,))


    # Handle sign ambiguity:  shortest path by checking dot product
    # If dot product is negative, the quaternions point in opposite directions
    # Flipping q_end to ensure we take the shortest path

    dot_product = np.dot(q_start, q_end)
    if dot_product < 0.0:
        q_end = -q_end
        dot_product = -dot_product

    # Clamp dot product to avoid numerical errors in arccos
    dot_product = np.clip(dot_product, -1.0, 1.0)

    # Calculate the angle between quaternions
    theta = np.arccos(dot_product)

    # Convert t to numpy array for vectorized operations
    t = np.atleast_1d(np.asarray(t, dtype=float))
```

```python
    is_scalar = t.shape == (1,)

    # Handle the case where quaternions are very close theta   0
    # Use linear interpolation to avoid division by zero
    sin_theta = np.sin(theta)

    if abs(sin_theta) < 1e-6:
        # Quaternions are nearly identical, use linear interpolation
        weights_start = 1.0 - t
        weights_end = t
    else:
        # Standard SLERP formula:
        # q(t) = [sin((1-t)* ) / sin( )] * q_start + [sin(t* ) / sin( )] * q_end

        weights_start = np.sin((1.0 - t) * theta) / sin_theta
        weights_end = np.sin(t * theta) / sin_theta

    # Perform interpolation
    # Shape handling:  weights can broadcast with quaternion
    result = (weights_start[:, np.newaxis] * q_start +
              weights_end[:, np.newaxis] * q_end)

    # Normalize result quaternions
    result = normalize_quaternion(result)

    # Return scalar if input was scalar
    if is_scalar:
        return result[0]
    return result
```

LERP Euler Implementation

[19]:
```python
def lerp_euler(euler_start: np.ndarray, euler_end: np.ndarray, t: Union[np.
 ↪ndarray, float]) -> np.ndarray:
    """
    Linear Interpolation (LERP) of Euler angles.

    Performs simple component-wise linear interpolation of Euler angles.
    Note: This is a naive approach and may not produce the most intuitive␣
 ↪rotations
    near gimbal lock configurations, but it's straightforward and works well for
    most practical cases with small angle differences.

    Args:
        euler_start: Starting Euler angles [angle1, angle2, angle3] in radians
        euler_end: Ending Euler angles [angle1, angle2, angle3] in radians
        t: Interpolation parameter(s), scalar or 1D array with values in [0, 1]
```

```
    Returns:
        Interpolated Euler angle(s) in radians
    """
    euler_start = np.asarray(euler_start, dtype=float).reshape(3,)
    euler_end = np.asarray(euler_end, dtype=float).reshape(3,)

    # Convert t to numpy array for vectorized operations
    t = np.atleast_1d(np.asarray(t, dtype=float))
    is_scalar = t.shape == (1,)

    # Component-wise linear interpolation
    # result(t) = (1-t) * euler_start + t * euler_end
    weights_start = 1.0 - t
    weights_end = t

    # interpolation: broadcast weights to multiply with euler angles
    result = (weights_start[:, np.newaxis] * euler_start +
              weights_end[:, np.newaxis] * euler_end)

    if is_scalar:
        return result[0]
    return result
```

Testing

[20]:
```
def run_notebook_tests(test_functions):
    """
    Args:
        test_functions: List of tuples (test_name, test_function)
    """
    passed = 0
    failed = 0
    errors = []

    for test_name, test_func in test_functions:
        try:
            test_func()
            print(f"PASSED: {test_name}")
            passed += 1
        except AssertionError as e:
            print(f"FAILED: {test_name}")
            print(f"Error: {str(e)}")
            failed += 1
            errors.append((test_name, str(e)))
        except Exception as e:
            print(f"ERROR: {test_name}")
            print(f"Error: {type(e).__name__}: {str(e)}")
            failed += 1
```

```
            errors.append((test_name, f"{type(e).__name__}: {str(e)}"))


    print(f"Results: {passed} passed, {failed} failed")


    if errors:
        print("\nFailed Tests Summary:")
        for name, error in errors:
            print(f"  - {name}: {error}")

    return passed, failed
```

SLERP Quaternion test

[21]:
```
def test_slerp_start():
    """Test: SLERP at t=0 returns start quaternion"""
    q1 = np.array([1.0, 0.0, 0.0, 0.0])
    q2 = axis_angle_quaternion("Z", np.pi / 2)
    result = slerp_quaternion(q1, q2, 0.0)
    assert np.allclose(result, q1), "SLERP at t=0 should equal q_start"

def test_slerp_end():
    """Test: SLERP at t=1 returns end quaternion"""
    q1 = np.array([1.0, 0.0, 0.0, 0.0])
    q2 = axis_angle_quaternion("Z", np.pi / 2)
    result = slerp_quaternion(q1, q2, 1.0)
    assert np.allclose(result, q2), "SLERP at t=1 should equal q_end"

def test_slerp_midpoint():
    """Test: SLERP midpoint is equidistant in rotation"""
    q1 = np.array([1.0, 0.0, 0.0, 0.0])
    q2 = axis_angle_quaternion("Z", np.pi / 2)
    q_mid = slerp_quaternion(q1, q2, 0.5)

    angle_start_to_mid = relative_rotation_angle(
        quaternion_to_rotation_matrix(q1),
        quaternion_to_rotation_matrix(q_mid)
    )
    angle_mid_to_end = relative_rotation_angle(
        quaternion_to_rotation_matrix(q_mid),
        quaternion_to_rotation_matrix(q2)
    )

    assert np.isclose(angle_start_to_mid, angle_mid_to_end, atol=1e-6), \
        f"Angles not equal: {np.degrees(angle_start_to_mid):.2f}° vs {np.
↪degrees(angle_mid_to_end):.2f}°"
```

```python
def test_slerp_array():
    """Test: SLERP with array t returns correct shape"""
    q1 = np.array([1.0, 0.0, 0.0, 0.0])
    q2 = axis_angle_quaternion("Z", np.pi / 2)
    t_values = np.array([0.0, 0.25, 0.5, 0.75, 1.0])
    result = slerp_quaternion(q1, q2, t_values)

    assert result.shape == (5, 4), f"Shape mismatch: {result.shape}"
    assert np.allclose(result[0], q1), "First result not q_start"
    assert np.allclose(result[-1], q2), "Last result not q_end"

def test_slerp_sign_ambiguity():
    """Test: SLERP handles sign ambiguity (shortest path)"""
    q_a = np.array([0.7071, 0.7071, 0.0, 0.0])
    q_b = np.array([-0.7071, -0.7071, 0.0, 0.0])
    result = slerp_quaternion(q_a, q_b, 0.5)

    dist_to_a = np.linalg.norm(result - q_a)
    dist_to_b = np.linalg.norm(result - q_b)

    assert np.min([dist_to_a, dist_to_b]) < 0.5, "Shortest path not taken"

def test_slerp_constant_velocity():
    """Test: SLERP maintains constant angular velocity"""
    q1 = np.array([1.0, 0.0, 0.0, 0.0])
    q2 = axis_angle_quaternion("Z", np.pi / 2)
    t_values = np.linspace(0, 1, 11)
    q_slerp = slerp_quaternion(q1, q2, t_values)

    angles = []
    for i in range(len(q_slerp) - 1):
        angle = relative_rotation_angle(
            quaternion_to_rotation_matrix(q_slerp[i]),
            quaternion_to_rotation_matrix(q_slerp[i + 1])
        )
        angles.append(angle)

    std_dev = np.std(angles)
    assert std_dev < 1e-5, f"Angular velocity not constant: std={std_dev}"

def test_slerp_small_angles():
    """Test: SLERP handles small angles without numerical issues"""
    q_start = np.array([1.0, 0.0, 0.0, 0.0])
    q_end = axis_angle_quaternion("X", 0.001)
    result = slerp_quaternion(q_start, q_end, 0.5)

    norm = np.linalg.norm(result)
```

```python
    assert np.isclose(norm, 1.0), f"Result not normalized: norm={norm}"

def test_slerp_normalized_output():
    """Test: All output quaternions are normalized"""
    q1 = np.array([1.0, 0.0, 0.0, 0.0])
    q2 = axis_angle_quaternion("Z", np.pi / 2)
    t_values = np.linspace(0, 1, 20)
    result = slerp_quaternion(q1, q2, t_values)

    norms = np.linalg.norm(result, axis=-1)
    assert np.allclose(norms, 1.0), f"Not all normalized: min={norms.min()},␣
↪max={norms.max()}"

def test_slerp_scalar_returns_1d():
    """Test: SLERP with scalar t returns 1D quaternion"""
    q1 = np.array([1.0, 0.0, 0.0, 0.0])
    q2 = axis_angle_quaternion("Z", np.pi / 2)
    result = slerp_quaternion(q1, q2, 0.5)

    assert result.shape == (4,), f"Shape mismatch: {result.shape}"
```

LERP Tests

[22]:
```python
def test_lerp_start():
    """Test: LERP at t=0 returns start angles"""
    euler_start = np.array([0.0, 0.0, 0.0])
    euler_end = np.array([np.pi/4, np.pi/6, np.pi/3])
    result = lerp_euler(euler_start, euler_end, 0.0)
    assert np.allclose(result, euler_start), "LERP at t=0 should equal␣
↪euler_start"

def test_lerp_end():
    """Test: LERP at t=1 returns end angles"""
    euler_start = np.array([0.0, 0.0, 0.0])
    euler_end = np.array([np.pi/4, np.pi/6, np.pi/3])
    result = lerp_euler(euler_start, euler_end, 1.0)
    assert np.allclose(result, euler_end), "LERP at t=1 should equal euler_end"

def test_lerp_midpoint():
    """Test: LERP midpoint is the average"""
    euler_start = np.array([0.0, 0.0, 0.0])
    euler_end = np.array([np.pi/4, np.pi/6, np.pi/3])
    result = lerp_euler(euler_start, euler_end, 0.5)
    expected = (euler_start + euler_end) / 2
    assert np.allclose(result, expected), "Midpoint should be average"

def test_lerp_array():
    """Test: LERP with array t returns correct shape"""
```

```python
    euler_start = np.array([0.0, 0.0, 0.0])
    euler_end = np.array([np.pi/4, np.pi/6, np.pi/3])
    t_values = np.array([0.0, 0.25, 0.5, 0.75, 1.0])
    result = lerp_euler(euler_start, euler_end, t_values)

    assert result.shape == (5, 3), f"Shape mismatch: {result.shape}"
    assert np.allclose(result[0], euler_start), "First result not euler_start"
    assert np.allclose(result[-1], euler_end), "Last result not euler_end"

def test_lerp_linearity():
    """Test: LERP produces linear progression"""
    euler_start = np.array([0.0, 0.0, 0.0])
    euler_end = np.array([1.0, 2.0, 3.0])
    t_values = np.array([0.0, 0.25, 0.5, 0.75, 1.0])
    result = lerp_euler(euler_start, euler_end, t_values)

    for i in range(3):
        expected = euler_start[i] + t_values * (euler_end[i] - euler_start[i])
        assert np.allclose(result[:, i], expected), f"Angle {i} not linear"

def test_lerp_scalar_returns_1d():
    """Test: LERP with scalar t returns 1D euler angles"""
    euler_start = np.array([0.0, 0.0, 0.0])
    euler_end = np.array([np.pi/4, np.pi/6, np.pi/3])
    result = lerp_euler(euler_start, euler_end, 0.5)
    assert result.shape == (3,), f"Shape mismatch: {result.shape}"

def test_lerp_different_ranges():
    """Test: LERP works with different angle ranges"""
    euler_start = np.array([-np.pi, 0.0, np.pi/2])
    euler_end = np.array([np.pi, np.pi/2, np.pi])
    result = lerp_euler(euler_start, euler_end, 0.5)
    expected = (euler_start + euler_end) / 2
    assert np.allclose(result, expected), "LERP with different ranges failed"
```

[23]:
```python
slerp_tests = [
    ("SLERP: t=0 returns start", test_slerp_start),
    ("SLERP: t=1 returns end", test_slerp_end),
    ("SLERP: Midpoint equidistant", test_slerp_midpoint),
    ("SLERP: Array input", test_slerp_array),
    ("SLERP: Sign ambiguity handling", test_slerp_sign_ambiguity),
    ("SLERP: Constant angular velocity", test_slerp_constant_velocity),
    ("SLERP: Small angles", test_slerp_small_angles),
    ("SLERP: Output normalized", test_slerp_normalized_output),
    ("SLERP: Scalar returns 1D", test_slerp_scalar_returns_1d),
]
```

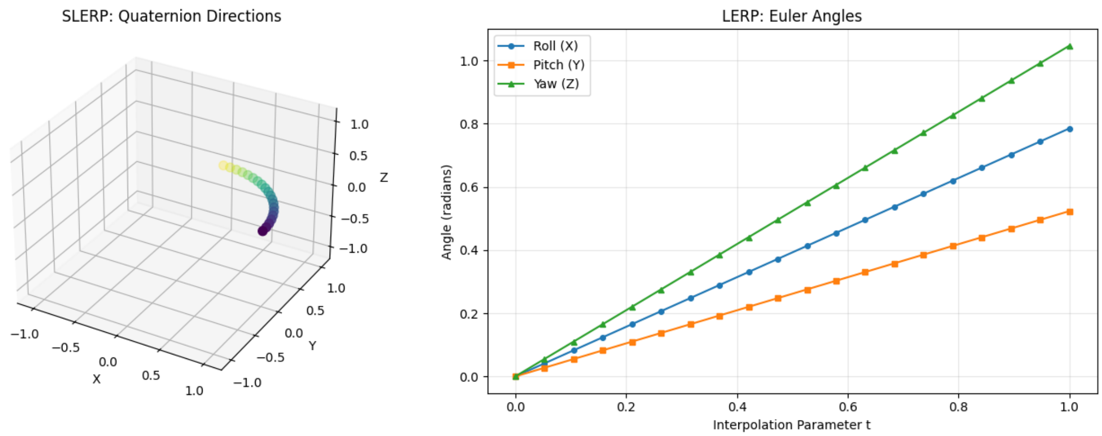
```
lerp_tests = [
    ("LERP: t=0 returns start", test_lerp_start),
    ("LERP: t=1 returns end", test_lerp_end),
    ("LERP: Midpoint is average", test_lerp_midpoint),
    ("LERP: Array input", test_lerp_array),
    ("LERP: Linearity", test_lerp_linearity),
    ("LERP: Scalar returns 1D", test_lerp_scalar_returns_1d),
    ("LERP: Different ranges", test_lerp_different_ranges),
]

all_tests = slerp_tests + lerp_tests

# Run tests
passed, failed = run_notebook_tests(all_tests)

print(f"Total Tests: {len(all_tests)}")
print(f"Passed: {passed}")
print(f"Failed: {failed}")
print(f"Success Rate: {100*passed/len(all_tests):.1f}%")
```

```
PASSED: SLERP: t=0 returns start
PASSED: SLERP: t=1 returns end
PASSED: SLERP: Midpoint equidistant
PASSED: SLERP: Array input
PASSED: SLERP: Sign ambiguity handling
PASSED: SLERP: Constant angular velocity
PASSED: SLERP: Small angles
PASSED: SLERP: Output normalized
PASSED: SLERP: Scalar returns 1D
PASSED: LERP: t=0 returns start
PASSED: LERP: t=1 returns end
PASSED: LERP: Midpoint is average
PASSED: LERP: Array input
PASSED: LERP: Linearity
PASSED: LERP: Scalar returns 1D
PASSED: LERP: Different ranges
Results: 16 passed, 0 failed
Total Tests: 16
Passed: 16
Failed: 0
Success Rate: 100.0%
```

`AI Caution Note:` test cases were developed with the assistance of `Claude Haiku 4.5` and were subsequently manually reviewed, adjusted, and refined to better align with the intended evaluation criteria.

[ ]:
```
# experiment :D  Visualize the interpolation results
import matplotlib.pyplot as plt
```

```
from mpl_toolkits.mplot3d import Axes3D

fig = plt.figure(figsize=(14, 5))

# SLERP visualization
ax1 = fig.add_subplot(121, projection='3d')
q1 = np.array([1.0, 0.0, 0.0, 0.0])
q2 = axis_angle_quaternion("Z", np.pi / 2)
t_values = np.linspace(0, 1, 20)
slerp_dirs = directions_from_quaternions(slerp_quaternion(q1, q2, t_values))

ax1.scatter(slerp_dirs[:, 0], slerp_dirs[:, 1], slerp_dirs[:, 2], c=t_values,␣
 ↪cmap='viridis', s=50)
ax1.set_xlabel('X')
ax1.set_ylabel('Y')
ax1.set_zlabel('Z')
ax1.set_title('SLERP: Quaternion Directions')
ax1.set_xlim([-1.2, 1.2])
ax1.set_ylim([-1.2, 1.2])
ax1.set_zlim([-1.2, 1.2])

# LERP visualization
ax2 = fig.add_subplot(122)
euler_start = np.array([0.0, 0.0, 0.0])
euler_end = np.array([np.pi/4, np.pi/6, np.pi/3])
t_values = np.linspace(0, 1, 20)
lerp_results = lerp_euler(euler_start, euler_end, t_values)

ax2.plot(t_values, lerp_results[:, 0], 'o-', label='Roll (X)', markersize=4)
ax2.plot(t_values, lerp_results[:, 1], 's-', label='Pitch (Y)', markersize=4)
ax2.plot(t_values, lerp_results[:, 2], '^-', label='Yaw (Z)', markersize=4)
ax2.set_xlabel('Interpolation Parameter t')
ax2.set_ylabel('Angle (radians)')
ax2.set_title('LERP: Euler Angles')
ax2.legend()
ax2.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```


SLERP: Quaternion Directions
X
Y
Z
−1.0
−0.5
0.0
0.5
1.0
LERP: Euler Angles
Roll (X)
Pitch (Y)
Yaw (Z)
Angle (radians)
Interpolation Parameter t
0.0
0.2
0.4
0.6
0.8
1.0

sheet_2_student_handout_m_Q3

May 16, 2026

1 Computer Animation - Assignment Sheet 2

[1]:
```
import numpy as np
from typing import Optional, Union
import matplotlib.pyplot as plt
from matplotlib.animation import FuncAnimation
from mpl_toolkits.mplot3d.art3d import Poly3DCollection

try:
    from IPython.display import HTML, display
except ImportError:
    HTML = None
    display = None

np.set_printoptions(precision=5, suppress=True)
EPS = 1e-10
```

[2]:
```
#for visuals
try:
    shell = get_ipython()
    try:
        shell.run_line_magic("matplotlib", "widget")
        print("Interactive backend enabled: %matplotlib widget")
    except Exception:
        shell.run_line_magic("matplotlib", "notebook")
        print("Interactive backend enabled: %matplotlib notebook")
except Exception:
    print("Interactive backend not enabled. For draggable 3D plots in␣
↪JupyterLab, install ipympl and run `%matplotlib widget`.")
```

```
Interactive backend enabled: %matplotlib widget
```

1.1 Provided helper functions from Sheet 1

[3]:
```
# ---------------------------------------------------------------------------
# Conventions used in this notebook
# ---------------------------------------------------------------------------
# Quaternions are represented as [w, x, y, z].
```

```
# Euler angles are represented in the order specified by `order`.
# Example: order="ZYX" means euler=[z_angle, y_angle, x_angle], and
# R = Rz(z_angle) @ Ry(y_angle) @ Rx(x_angle).
# All six Tait-Bryan Euler orders are supported:
# "XYZ", "XZY", "YXZ", "YZX", "ZXY", "ZYX".
# -----------------------------------------------------------------------------

ORDER = "ZYX"

def normalize_vector(v: np.ndarray, axis: int = -1, eps: float = EPS) -> np.
↪ndarray:
    """Normalize vectors along the given axis."""
    v = np.asarray(v, dtype=float)
    n = np.linalg.norm(v, axis=axis, keepdims=True)
    if np.any(n < eps):
        raise ValueError("Cannot normalize a vector with norm close to zero.")
    return v / n


def normalize_quaternion(q: np.ndarray, eps: float = EPS) -> np.ndarray:
    """Normalize one quaternion of shape (4,) or an array of quaternions (...,␣
↪4)."""
    q = np.asarray(q, dtype=float)
    if q.shape[-1] != 4:
        raise ValueError("Quaternions must have shape (..., 4).")
    n = np.linalg.norm(q, axis=-1, keepdims=True)
    if np.any(n < eps):
        raise ValueError("Cannot normalize a quaternion with norm close to zero.
↪")
    return q / n


def quaternion_conjugate(q: np.ndarray) -> np.ndarray:
    """Return the conjugate of q = [w, x, y, z]."""
    q = np.asarray(q, dtype=float)
    if q.shape[-1] != 4:
        raise ValueError("Quaternions must have shape (..., 4).")
    out = q.copy()
    out[..., 1:] *= -1.0
    return out


def quaternion_norm(q: np.ndarray) -> np.ndarray:
    """Return the Euclidean norm of one quaternion or a batch of quaternions."""
    q = np.asarray(q, dtype=float)
    if q.shape[-1] != 4:
        raise ValueError("Quaternions must have shape (..., 4).")
```

```python
    return np.linalg.norm(q, axis=-1)


def quaternion_multiply(q_left: np.ndarray, q_right: np.ndarray) -> np.ndarray:
    """Hamilton product q_left * q_right for quaternions [w, x, y, z]."""
    q_left = np.asarray(q_left, dtype=float)
    q_right = np.asarray(q_right, dtype=float)
    if q_left.shape[-1] != 4 or q_right.shape[-1] != 4:
        raise ValueError("Both inputs must have shape (..., 4).")

    w1, x1, y1, z1 = np.moveaxis(q_left, -1, 0)
    w2, x2, y2, z2 = np.moveaxis(q_right, -1, 0)

    return np.stack([
        w1*w2 - x1*x2 - y1*y2 - z1*z2,
        w1*x2 + x1*w2 + y1*z2 - z1*y2,
        w1*y2 - x1*z2 + y1*w2 + z1*x2,
        w1*z2 + x1*y2 - y1*x2 + z1*w2,
    ], axis=-1)


def rotation_matrix_x(angle: float) -> np.ndarray:
    """Rotation matrix for a rotation about the x-axis."""
    c, s = np.cos(angle), np.sin(angle)
    return np.array([[1.0, 0.0, 0.0], [0.0, c, -s], [0.0, s, c]])


def rotation_matrix_y(angle: float) -> np.ndarray:
    """Rotation matrix for a rotation about the y-axis."""
    c, s = np.cos(angle), np.sin(angle)
    return np.array([[c, 0.0, s], [0.0, 1.0, 0.0], [-s, 0.0, c]])


def rotation_matrix_z(angle: float) -> np.ndarray:
    """Rotation matrix for a rotation about the z-axis."""
    c, s = np.cos(angle), np.sin(angle)
    return np.array([[c, -s, 0.0], [s, c, 0.0], [0.0, 0.0, 1.0]])


def axis_rotation_matrix(axis: str, angle: float) -> np.ndarray:
    """Return a basic rotation matrix for the selected axis."""
    axis = axis.upper()
    if axis == "X":
        return rotation_matrix_x(angle)
    if axis == "Y":
        return rotation_matrix_y(angle)
    if axis == "Z":
```

```python
        return rotation_matrix_z(angle)
    raise ValueError("axis must be 'X', 'Y', or 'Z'.")


def axis_angle_quaternion(axis: str, angle: float) -> np.ndarray:
    """Quaternion [w, x, y, z] for a rotation around one coordinate axis."""
    half = angle / 2.0
    c, s = np.cos(half), np.sin(half)
    axis = axis.upper()
    if axis == "X":
        return np.array([c, s, 0.0, 0.0])
    if axis == "Y":
        return np.array([c, 0.0, s, 0.0])
    if axis == "Z":
        return np.array([c, 0.0, 0.0, s])
    raise ValueError("axis must be 'X', 'Y', or 'Z'.")


def euler_to_rotation_matrix(euler: np.ndarray, order: str = ORDER) -> np.
 ↪ndarray:
    """
    Convert Euler angles to a rotation matrix.

    The angles are interpreted in the same order as the order string.
    Example: order="ZYX", euler=[z_angle, y_angle, x_angle].
    """
    euler = np.asarray(euler, dtype=float).reshape(3,)

    R = np.eye(3)
    for axis, angle in zip(order, euler):
        R = R @ axis_rotation_matrix(axis, angle)
    return R


def euler_to_quaternion(euler: np.ndarray, order: str = ORDER) -> np.ndarray:
    """
    Convert Euler angles to a unit quaternion [w, x, y, z].

    The angles are interpreted in the same order as the order string.
    Example: order="ZYX", euler=[z_angle, y_angle, x_angle].
    """
    euler = np.asarray(euler, dtype=float).reshape(3,)

    q = np.array([1.0, 0.0, 0.0, 0.0])
    for axis, angle in zip(order, euler):
        q = quaternion_multiply(q, axis_angle_quaternion(axis, angle))
    return normalize_quaternion(q)
```

```python
def quaternion_to_rotation_matrix(q: np.ndarray) -> np.ndarray:
    """Convert q = [w, x, y, z] to a 3 x 3 rotation matrix."""
    q = normalize_quaternion(np.asarray(q, dtype=float).reshape(4,))
    w, x, y, z = q
    return np.array([
        [1 - 2*(y*y + z*z), 2*(x*y - w*z),     2*(x*z + w*y)],
        [2*(x*y + w*z),     1 - 2*(x*x + z*z), 2*(y*z - w*x)],
        [2*(x*z - w*y),     2*(y*z + w*x),     1 - 2*(x*x + y*y)],
    ], dtype=float)


def rotation_matrix_to_quaternion(R: np.ndarray) -> np.ndarray:
    """Convert a 3 x 3 rotation matrix to a unit quaternion [w, x, y, z]."""
    R = np.asarray(R, dtype=float).reshape(3, 3)
    trace = np.trace(R)

    if trace > 0.0:
        s = 2.0 * np.sqrt(trace + 1.0)
        w = 0.25 * s
        x = (R[2, 1] - R[1, 2]) / s
        y = (R[0, 2] - R[2, 0]) / s
        z = (R[1, 0] - R[0, 1]) / s
    elif R[0, 0] > R[1, 1] and R[0, 0] > R[2, 2]:
        s = 2.0 * np.sqrt(1.0 + R[0, 0] - R[1, 1] - R[2, 2])
        w = (R[2, 1] - R[1, 2]) / s
        x = 0.25 * s
        y = (R[0, 1] + R[1, 0]) / s
        z = (R[0, 2] + R[2, 0]) / s
    elif R[1, 1] > R[2, 2]:
        s = 2.0 * np.sqrt(1.0 + R[1, 1] - R[0, 0] - R[2, 2])
        w = (R[0, 2] - R[2, 0]) / s
        x = (R[0, 1] + R[1, 0]) / s
        y = 0.25 * s
        z = (R[1, 2] + R[2, 1]) / s
    else:
        s = 2.0 * np.sqrt(1.0 + R[2, 2] - R[0, 0] - R[1, 1])
        w = (R[1, 0] - R[0, 1]) / s
        x = (R[0, 2] + R[2, 0]) / s
        y = (R[1, 2] + R[2, 1]) / s
        z = 0.25 * s

    return normalize_quaternion(np.array([w, x, y, z]))
```

```python
def rotation_matrix_to_euler(R: np.ndarray, order: str = ORDER, eps: float = 1e-8) -> np.ndarray:
    """
    Convert a rotation matrix to Euler angles for any of the six supported orders.

    Returns angles in the same order as `order`.
    Example: order="ZYX" returns [z_angle, y_angle, x_angle].
    In gimbal-lock cases, the last angle is set to zero.
    """
    R = np.asarray(R, dtype=float).reshape(3, 3)

    if order == "XYZ":
        sb = np.clip(R[0, 2], -1.0, 1.0)
        b = np.arcsin(sb)
        if abs(abs(sb) - 1.0) < eps:
            c = 0.0
            a = np.arctan2(R[1, 0], R[1, 1]) if sb > 0 else np.arctan2(-R[1, 0], R[1, 1])
        else:
            a = np.arctan2(-R[1, 2], R[2, 2])
            c = np.arctan2(-R[0, 1], R[0, 0])

    elif order == "XZY":
        sb = np.clip(-R[0, 1], -1.0, 1.0)
        b = np.arcsin(sb)
        if abs(abs(sb) - 1.0) < eps:
            c = 0.0
            a = np.arctan2(-R[1, 2], R[1, 0]) if sb > 0 else np.arctan2(-R[1, 2], -R[1, 0])
        else:
            a = np.arctan2(R[2, 1], R[1, 1])
            c = np.arctan2(R[0, 2], R[0, 0])

    elif order == "YXZ":
        sb = np.clip(-R[1, 2], -1.0, 1.0)
        b = np.arcsin(sb)
        if abs(abs(sb) - 1.0) < eps:
            c = 0.0
            a = np.arctan2(-R[2, 0], R[0, 0])
        else:
            a = np.arctan2(R[0, 2], R[2, 2])
            c = np.arctan2(R[1, 0], R[1, 1])

    elif order == "YZX":
        sb = np.clip(R[1, 0], -1.0, 1.0)
        b = np.arcsin(sb)
```

```
        if abs(abs(sb) - 1.0) < eps:
            c = 0.0
            a = np.arctan2(R[0, 2], R[2, 2])
        else:
            a = np.arctan2(-R[2, 0], R[0, 0])
            c = np.arctan2(-R[1, 2], R[1, 1])

    elif order == "ZXY":
        sb = np.clip(R[2, 1], -1.0, 1.0)
        b = np.arcsin(sb)
        if abs(abs(sb) - 1.0) < eps:
            c = 0.0
            a = np.arctan2(R[1, 0], R[0, 0])
        else:
            a = np.arctan2(-R[0, 1], R[1, 1])
            c = np.arctan2(-R[2, 0], R[2, 2])

    elif order == "ZYX":
        sb = np.clip(-R[2, 0], -1.0, 1.0)
        b = np.arcsin(sb)
        if abs(abs(sb) - 1.0) < eps:
            c = 0.0
            a = np.arctan2(R[1, 2], R[0, 2]) if sb > 0 else np.arctan2(-R[1,
↪2], -R[0, 2])
        else:
            a = np.arctan2(R[1, 0], R[0, 0])
            c = np.arctan2(R[2, 1], R[2, 2])

    return np.array([a, b, c])


def quaternion_to_euler(q: np.ndarray, order: str = ORDER) -> np.ndarray:
    """Convert a quaternion [w, x, y, z] to Euler angles in the requested order.
↪"""
    return rotation_matrix_to_euler(quaternion_to_rotation_matrix(q),
↪order=order)


def rotation_direction_from_matrix(R: np.ndarray) -> np.ndarray:
    """Direction of the local +x axis after applying R."""
    return R @ np.array([1.0, 0.0, 0.0])


def directions_from_quaternions(quaternions: np.ndarray) -> np.ndarray:
    """Map a sequence of quaternions to the corresponding local +x directions.
↪"""
```

```
    return np.
↪array([rotation_direction_from_matrix(quaternion_to_rotation_matrix(q)) for␣
↪q in quaternions])


def directions_from_eulers(eulers: np.ndarray, order: str = ORDER) -> np.
↪ndarray:
    """Map a sequence of Euler angles to the corresponding local +x directions.
↪"""
    return np.array([rotation_direction_from_matrix(euler_to_rotation_matrix(e,␣
↪order=order)) for e in eulers])


def relative_rotation_angle(R0: np.ndarray, R1: np.ndarray) -> float:
    """Return the angle of the relative rotation R0.T @ R1 in radians."""
    R_rel = R0.T @ R1
    c = (np.trace(R_rel) - 1.0) / 2.0
    return float(np.arccos(np.clip(c, -1.0, 1.0)))
```

1.2 Provided mesh and plotting helpers for Assignment 3

[4]:
```
ARROW_LENGTH = 0.45
SPHERE_RADIUS = 1.0


def create_sphere_mesh(radius: float = SPHERE_RADIUS, n_lat: int = 24, n_lon:␣
↪int = 48):
    """Return x, y, z arrays for plotting a sphere with Matplotlib."""
    u = np.linspace(0.0, 2.0*np.pi, n_lon)
    v = np.linspace(0.0, np.pi, n_lat)
    x = radius * np.outer(np.cos(u), np.sin(v))
    y = radius * np.outer(np.sin(u), np.sin(v))
    z = radius * np.outer(np.ones_like(u), np.cos(v))
    return x, y, z


def create_arrow_mesh(length: float = ARROW_LENGTH,
                      shaft_length: float = 0.28,
                      shaft_radius: float = 0.025,
                      head_radius: float = 0.085,
                      n_segments: int = 16):
    """
    Create a simple arrow mesh whose tip points along the local negative x-axis.

    Returns
    -------
    vertices : ndarray, shape (n_vertices, 3)
```

```
    faces : list[list[int]]
        Indices into vertices. Faces can be triangles or quads.
    """
    theta = np.linspace(0.0, 2.0*np.pi, n_segments, endpoint=False)

    base = np.column_stack([
        np.zeros(n_segments),
        shaft_radius * np.cos(theta),
        shaft_radius * np.sin(theta),
    ])
    shaft_end = np.column_stack([
        -np.full(n_segments, shaft_length),
        shaft_radius * np.cos(theta),
        shaft_radius * np.sin(theta),
    ])
    head_base = np.column_stack([
        -np.full(n_segments, shaft_length),
        head_radius * np.cos(theta),
        head_radius * np.sin(theta),
    ])
    tip = np.array([[-length, 0.0, 0.0]])
    base_center = np.array([[0.0, 0.0, 0.0]])

    vertices = np.vstack([base, shaft_end, head_base, tip, base_center])
    tip_idx = 3 * n_segments
    base_center_idx = 3 * n_segments + 1

    faces = []
    for i in range(n_segments):
        j = (i + 1) % n_segments
        faces.append([i, j, n_segments + j, n_segments + i])
        faces.append([n_segments + i, n_segments + j, 2*n_segments + j,␣
↪2*n_segments + i])
        faces.append([2*n_segments + i, 2*n_segments + j, tip_idx])
        faces.append([base_center_idx, j, i])

    return vertices, faces


def transform_vertices(vertices: np.ndarray,
                       R: np.ndarray,
                       offset: Optional[np.ndarray] = None,
                       scale: float = 1.0) -> np.ndarray:
    """Apply scale, rotation, and translation to vertices."""
    vertices = np.asarray(vertices, dtype=float)
    R = np.asarray(R, dtype=float).reshape(3, 3)
    if offset is None:
```

```
        offset = np.zeros(3)
    offset = np.asarray(offset, dtype=float).reshape(3,)
    return scale * (vertices @ R.T) + offset


def add_arrow(ax,
              vertices: np.ndarray,
              faces: list,
              R: np.ndarray,
              offset: np.ndarray,
              color: str = "tab:blue",
              alpha: float = 0.55):
    """Add one transformed arrow mesh to a 3D axis."""
    transformed = transform_vertices(vertices, R, offset=offset)
    poly = [[transformed[idx] for idx in face] for face in faces]
    collection = Poly3DCollection(poly, alpha=alpha, linewidths=0.4)
    collection.set_facecolor(color)
    collection.set_edgecolor("black")
    ax.add_collection3d(collection)
    return collection


def inward_arrow_offset(direction: np.ndarray,
                        sphere_radius: float = SPHERE_RADIUS,
                        arrow_length: float = ARROW_LENGTH) -> np.ndarray:
    """
    Place the arrow base outside the sphere so the arrow tip lies on the sphere.
    """
    direction = normalize_vector(np.asarray(direction, dtype=float).reshape(3,))
    return (sphere_radius + arrow_length) * direction


def set_axes_equal(ax, limit: float = 1.75):
    """Use equal limits on all three 3D axes."""
    ax.set_xlim(-limit, limit)
    ax.set_ylim(-limit, limit)
    ax.set_zlim(-limit, limit)
    ax.set_box_aspect((1, 1, 1))
    ax.set_xlabel("x")
    ax.set_ylabel("y")
    ax.set_zlabel("z")


def plot_interpolation_paths(q_sequence: np.ndarray,
                             euler_sequence: np.ndarray,
                             order: str = ORDER,
                             sample_indices: Optional[np.ndarray] = None,
```

```
                              title: str = "Quaternion SLERP vs. Euler LERP"):
    """Plot the orientation paths of the local +x axis on the sphere."""
    q_sequence = np.asarray(q_sequence, dtype=float)
    euler_sequence = np.asarray(euler_sequence, dtype=float)

    q_dirs = directions_from_quaternions(q_sequence)
    e_dirs = directions_from_eulers(euler_sequence, order=order)

    vertices, faces = create_arrow_mesh()
    sphere_x, sphere_y, sphere_z = create_sphere_mesh()

    if sample_indices is None:
        sample_indices = np.linspace(0, len(q_sequence) - 1, 7, dtype=int)

    fig = plt.figure(figsize=(8, 7))
    ax = fig.add_subplot(111, projection="3d")
    ax.plot_wireframe(sphere_x, sphere_y, sphere_z, linewidth=0.25, alpha=0.25)
    ax.plot(q_dirs[:, 0], q_dirs[:, 1], q_dirs[:, 2], color="tab:blue",␣
↪linewidth=2.0, label="quaternion SLERP")
    ax.plot(e_dirs[:, 0], e_dirs[:, 1], e_dirs[:, 2], color="tab:green",␣
↪linewidth=2.0, label="Euler LERP")
    ax.scatter(q_dirs[[0, -1], 0], q_dirs[[0, -1], 1], q_dirs[[0, -1], 2],␣
↪color="black", s=35)

    for idx in sample_indices:
        q_R = quaternion_to_rotation_matrix(q_sequence[idx])
        e_R = euler_to_rotation_matrix(euler_sequence[idx], order=order)
        add_arrow(ax, vertices, faces, q_R,␣
↪offset=inward_arrow_offset(q_dirs[idx]), color="tab:blue", alpha=0.35)
        add_arrow(ax, vertices, faces, e_R,␣
↪offset=inward_arrow_offset(e_dirs[idx]), color="tab:green", alpha=0.35)

    set_axes_equal(ax)
    ax.set_title(title + f"   (Euler order: {order})")
    ax.legend(loc="upper left")
    return fig, ax


def plot_angular_distance(t_values: np.ndarray,
                          q_sequence: np.ndarray,
                          euler_sequence: np.ndarray,
                          euler_start: np.ndarray,
                          order: str = ORDER):
    """Plot angular distance from the start orientation over interpolation time.
↪"""
    t_values = np.asarray(t_values, dtype=float)
```

```
    R0 = euler_to_rotation_matrix(euler_start, order=order)

    q_angles = np.array([
        relative_rotation_angle(R0, quaternion_to_rotation_matrix(q))
        for q in q_sequence
    ])
    e_angles = np.array([
        relative_rotation_angle(R0, euler_to_rotation_matrix(e, order=order))
        for e in euler_sequence
    ])

    fig, ax = plt.subplots(figsize=(7, 4))
    ax.plot(t_values, np.rad2deg(q_angles), label="quaternion SLERP")
    ax.plot(t_values, np.rad2deg(e_angles), label="Euler LERP")
    ax.set_xlabel("t")
    ax.set_ylabel("angular distance from start [degrees]")
    ax.set_title("Interpolation progress over time")
    ax.grid(True, alpha=0.3)
    ax.legend()
    return fig, ax
```

1.3 Assignment 3: Animation

[5]:
```
def animate_interpolation(q_sequence: np.ndarray,
                          euler_sequence: np.ndarray,
                          order: str = ORDER,
                          interval: int = 70):

    q_dirs = directions_from_quaternions(q_sequence)
    e_dirs = directions_from_eulers(euler_sequence)

    vertices, faces = create_arrow_mesh()
    sphere_x, sphere_y, sphere_z = create_sphere_mesh()

    fig = plt.figure(figsize=(8, 7))
    ax = fig.add_subplot(111, projection="3d")

    # sphere wireframe- static
    ax.plot_wireframe(sphere_x, sphere_y, sphere_z, linewidth=0.25, alpha=0.25)

    ax.plot(q_dirs[:, 0], q_dirs[:, 1], q_dirs[:, 2],
            color="tab:blue", linewidth=1.5, alpha=0.35, label="quaternion␣
↪SLERP")
    ax.plot(e_dirs[:, 0], e_dirs[:, 1], e_dirs[:, 2],
            color="tab:green", linewidth=1.5, alpha=0.35, label="Euler LERP")

    set_axes_equal(ax)
```

```
    ax.legend(loc="upper left")
    ax.set_title(f"Quaternion SLERP vs. Euler LERP   (Euler order: {order})")

    q_coll = [None]
    e_coll = [None]

    # dynamic part
    def update(frame):
        if q_coll[0] is not None:
            q_coll[0].remove()
        if e_coll[0] is not None:
            e_coll[0].remove()

        q_R = quaternion_to_rotation_matrix(q_sequence[frame])
        e_R = euler_to_rotation_matrix(euler_sequence[frame], order=order)

        q_coll[0] = add_arrow(ax, vertices, faces, q_R,
                              offset=inward_arrow_offset(q_dirs[frame]),
                              color="tab:blue", alpha=0.8)
        e_coll[0] = add_arrow(ax, vertices, faces, e_R,
                              offset=inward_arrow_offset(e_dirs[frame]),
                              color="tab:green", alpha=0.8)
        return q_coll[0], e_coll[0]

    return FuncAnimation(fig, update, frames=len(q_sequence),␣
↪interval=interval, blit=False)


    # raise NotImplementedError("TODO")
```

1.4 Assignment 2.1: Spherical linear interpolation of quaternions

[6]:
```
def slerp_quaternion(q_start: np.ndarray, q_end: np.ndarray, t: Union[np.
↪ndarray, float]) -> np.ndarray:
    """
    Note:
    Spherical Linear Interpolation (SLERP) between two quaternions.
     performs smooth interpolation along the shortest path on the unit␣
↪quaternion sphere.

    Args:
        q_start: Starting quaternion [w, x, y, z]
        q_end: Ending quaternion [w, x, y, z]
        t: Interpolation parameter(s), scalar or 1D array with values in [0, 1]␣
↪- interpolation factors? 0 = q_start, 1 = q_end

    Returns:
```

```
        Interpolated quaternions with same shape as t (scalar or array)
    """
    #  inputs are numpy arrays and normalize
    q_start = normalize_quaternion(np.asarray(q_start, dtype=float).reshape(4,))
    q_end = normalize_quaternion(np.asarray(q_end, dtype=float).reshape(4,))


    # Handle sign ambiguity:  shortest path by checking dot product
    # If dot product is negative, the quaternions point in opposite directions
    # Flipping q_end to ensure we take the shortest path

    dot_product = np.dot(q_start, q_end)
    if dot_product < 0.0:
        q_end = -q_end
        dot_product = -dot_product

    # Clamp dot product to avoid numerical errors in arccos
    dot_product = np.clip(dot_product, -1.0, 1.0)

    # Calculate the angle between quaternions
    theta = np.arccos(dot_product)

    # Convert t to numpy array for vectorized operations
    t = np.atleast_1d(np.asarray(t, dtype=float))
    is_scalar = t.shape == (1,)

    # Handle the case where quaternions are very close theta   0
    # Use linear interpolation to avoid division by zero
    sin_theta = np.sin(theta)

    if abs(sin_theta) < 1e-6:
        # Quaternions are nearly identical, use linear interpolation
        weights_start = 1.0 - t
        weights_end = t
    else:
        # Standard SLERP formula:
        # q(t) = [sin((1-t)* ) / sin( )] * q_start + [sin(t* ) / sin( )] * q_end

        weights_start = np.sin((1.0 - t) * theta) / sin_theta
        weights_end = np.sin(t * theta) / sin_theta

    # Perform interpolation
    # Shape handling:  weights can broadcast with quaternion
    result = (weights_start[:, np.newaxis] * q_start +
              weights_end[:, np.newaxis] * q_end)

    # Normalize result quaternions
```

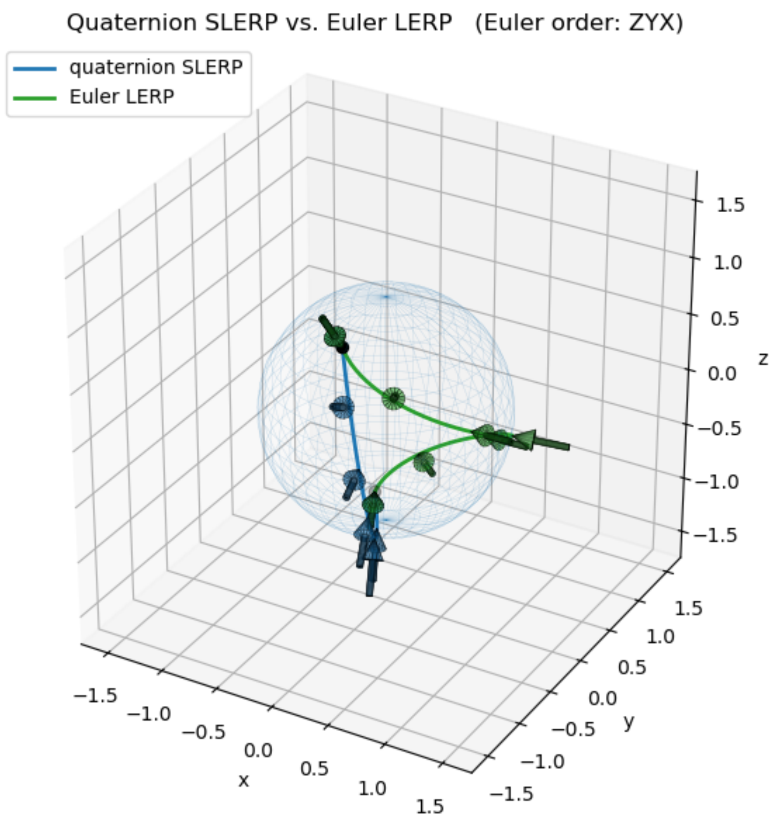
```
    result = normalize_quaternion(result)

    # Return scalar if input was scalar
    if is_scalar:
        return result[0]
    return result
```

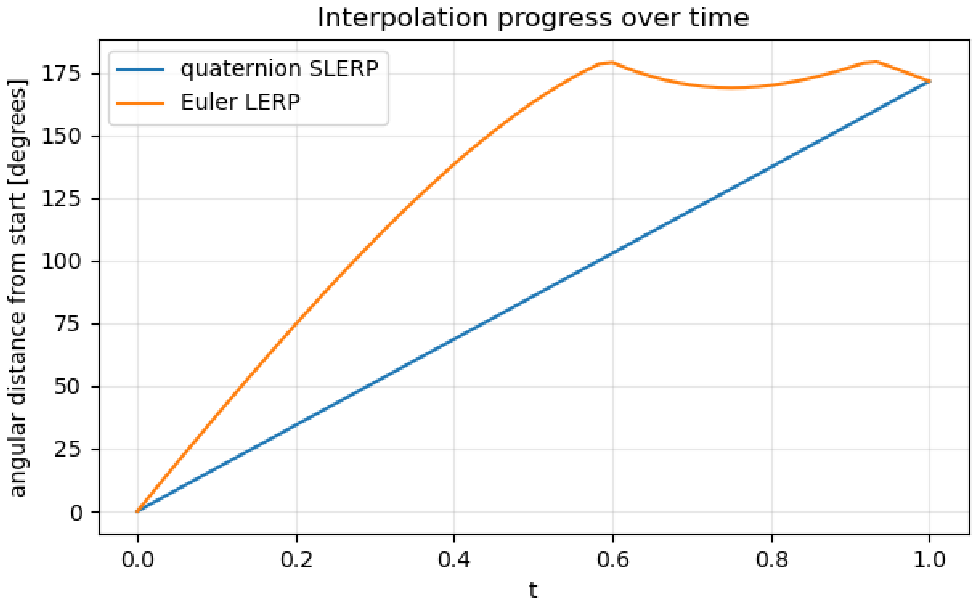
1.5 Assignment 2.2: Linear interpolation of Euler angles

[7]:
```
def lerp_euler(euler_start: np.ndarray, euler_end: np.ndarray, t: Union[np.
 ↪ndarray, float]) -> np.ndarray:
    """
    Linear Interpolation (LERP) of Euler angles.

    Performs simple component-wise linear interpolation of Euler angles.
    Note: This is a naive approach and may not produce the most intuitive␣
 ↪rotations
    near gimbal lock configurations, but it's straightforward and works well for
    most practical cases with small angle differences.

    Args:
        euler_start: Starting Euler angles [angle1, angle2, angle3] in radians
        euler_end: Ending Euler angles [angle1, angle2, angle3] in radians
        t: Interpolation parameter(s), scalar or 1D array with values in [0, 1]

    Returns:
        Interpolated Euler angle(s) in radians
    """
    euler_start = np.asarray(euler_start, dtype=float).reshape(3,)
    euler_end = np.asarray(euler_end, dtype=float).reshape(3,)

    # Convert t to numpy array for vectorized operations
    t = np.atleast_1d(np.asarray(t, dtype=float))
    is_scalar = t.shape == (1,)

    # Component-wise linear interpolation
    # result(t) = (1-t) * euler_start + t * euler_end
    weights_start = 1.0 - t
    weights_end = t

    # interpolation: broadcast weights to multiply with euler angles
    result = (weights_start[:, np.newaxis] * euler_start +
              weights_end[:, np.newaxis] * euler_end)

    if is_scalar:
        return result[0]
    return result
```

1.6 Assignment 3: Illustration and comparison

[8]:
```
# Example setup for Assignment 3.

ORDER = "ZYX"
t_values = np.linspace(0.0, 1.0, 61)

euler_start = np.deg2rad([-95.0, -55.0, 10.0])
euler_end = np.deg2rad([135.0, 70.0, 165.0])

q_start = euler_to_quaternion(euler_start, order=ORDER)
q_end = euler_to_quaternion(euler_end, order=ORDER)

# TODO
q_sequence = slerp_quaternion(q_start, q_end, t_values)
euler_sequence = lerp_euler(euler_start, euler_end, t_values)

fig, ax = plot_interpolation_paths(q_sequence, euler_sequence, order=ORDER)
plt.show()

fig2, ax2 = plot_angular_distance(t_values, q_sequence, euler_sequence,
    euler_start, order=ORDER)
plt.show()

anim = animate_interpolation(q_sequence, euler_sequence, order=ORDER,
    interval=70)
if HTML is not None:
    display(HTML(anim.to_jshtml()))
```


Quaternion SLERP vs. Euler LERP (Euler order: ZYX)
quaternion SLERP
Euler LERP
x
y
z
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5




```
&lt;IPython.core.display.HTML object&gt;
```

1.6.1 Observations:

The SLERP path traces a geodesic arc (the shortest path between two points on a sphere) — it's a smooth, consistent curve.

The Euler LERP path takes a different, longer route as it curves away from the geodesic because blending three angles independently doesn't correspond to a straight rotation path.

SLERP moves at a constant angular velocity, as the angular distance from the plot shows a straight, this is not the case with Euler LERP.